

Design, Analysis and Simulation of a Mechanical Watch Going Train with a Swiss Lever Escapement

Joshua John Leo Valan Johnson

Department of Instrumentation and Control Engineering

National Institute of Technology Tiruchirappalli

Tiruchirappalli, India

Video, complete train running: https://www.youtube.com/watch?v=dhshj_wPEz8

Video, escapement close-up: https://youtu.be/M3oqJB_sfAk

Full technical report (81 pages): <https://drive.google.com/file/d/1B5jwz9EtaVPRjgJI0gEpOFXx94d87E19/view?usp=sharing>

Abstract—This paper presents the design of the basic mechanism of a mechanical watch, from the mainspring barrel through a three-wheel going train and a Swiss lever escapement to the balance and hairspring. The mechanism comprises thirteen unique parts, modelled in SolidWorks from one file of 393 linked equations. A train ratio of 6720 gives 28 800 beats per hour. All teeth follow the ogival NIHS 20-30 form, reconstructed as constrained sketches that match generated profiles to better than a nanometre on the working flanks. A contact analysis with a 1000-sample Monte Carlo study per mesh set the width of the barrel teeth and the centre-distance tolerances. Checking each element from first principles exposed nine defects, of which the most serious was a barrel that would have held the balance above its running limit for 19 hours instead of 40. A rigid-body contact simulation of the complete train then gave a balance period of 0.2498 s against a design value of 0.2500 s and exact transmission ratios. The full derivations are given in a companion technical report.

Index Terms—mechanical watch, going train, Swiss lever escapement, ogival gearing, NIHS 20-30, multibody contact simulation, tolerance analysis

I. INTRODUCTION

A mechanical watch stores about 0.15J in a coiled spring and releases it over two days through a mechanism that runs on about a microwatt. This project designs that mechanism from its requirements: the barrel and mainspring, the going train, a Swiss lever escapement, and the balance with its hairspring. The concept and the starting requirements, 28 800 beats per hour and about 40 hours of running, come from Ciechanowski's illustrated article [1], which served as reading and design inspiration only. None of its geometry, models or code is reused: every part was dimensioned from the calculations summarised here and modelled from an empty file.

The aim was not a wearable movement but a design in which every dimension has a derivation, followed by a check that the result works. The contributions are:

- a fully parametric model in which 192 inputs drive 201 derived dimensions and check values;
- a reconstruction of the NIHS 20-30 tooth form as constrained geometry, verified against independently generated profiles;

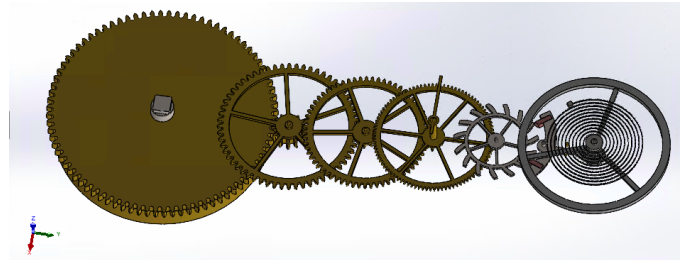


Fig. 1. The complete mechanism, from the mainspring barrel (left) through the going train and escapement to the balance (right): thirteen unique parts on one line of centres.

- a contact and tolerance study that sized the tooth widths and the centre-distance tolerances;
- a machine-element verification of every load-carrying feature, which exposed nine defects and changed the model in seven places;
- a contact simulation of the complete train, with settings derived from the physics of the impacts.

Bearings, plates, bridges and shock protection are outside the model, and nothing has been manufactured. Section XI states what this leaves open. This paper is a condensed account; the companion report [2] gives every derivation and repeats each calculation with the numbers substituted line by line.

Table I lists the requirements. The rate, the reserve and the display were given at the outset; the strength and life targets were set during the analysis so that every check had a stated pass criterion. Below about 150° of amplitude a watch still runs but its rate is poor and it is close to stopping; above about 315° the roller jewel strikes the outside of the fork.

II. MECHANISM AND PARAMETRIC MODEL

The mainspring turns a toothed barrel. Three wheels, each driving a pinion on the next arbor, multiply its speed 420 times to the fourth wheel, which turns once per minute, and a further 16 times to the escape wheel. The escapement locks the train and releases it one tooth for every oscillation of the balance, giving the balance a small impulse at each release.

TABLE I
DESIGN REQUIREMENTS

Requirement	Value
Rate	28 800 beats per hour, a 4Hz balance
Power reserve	at least 40 hours, with an amplitude of at least 150° at 40 hours
Amplitude	at most 315° at full wind; at least 200° after 24 hours
Time display	fourth wheel once per minute; second wheel once per hour
Tooth form	NIHS 20-30 ogival profiles on every mesh
Strength	safety factors of 2.0 on yield in running, 1.5 to 2.0 on yield under a 5000g shock, 1.5 in fatigue and 1.2 on first yield in contact
Service life	ten years, 1.26×10^9 oscillations of the balance

TABLE II
MATERIALS AND THE PROPERTIES USED IN THE CALCULATIONS (MPa)

Parts	Material	Properties used
Wheels, barrel	Brass CuZn39Pb2, hard	$R_{p0.2}$ 400, R_m 550
Pinions, arbors, escapement	Hardened carbon steel	$R_{p0.2}$ 1600, R_m 2000
Balance	Copper-beryllium	$R_{p0.2}$ 1000
Stones, roller jewel	Synthetic ruby	E 400 000
Hairspring	Nivarox	E 190 000
Mainspring	Nivaflex 45/5	E 220 000, $R_{p0.2}$ 2475

Fig. 3 shows the path of the energy. The thirteen parts lie on a single line of centres, each mesh on its own level 0.5mm above the last, so that the whole energy path is visible (Fig. 1). Every centre distance has its theoretical value $a = m(Z + p)/2$.

The parts are shown in Fig. 2 and their materials in Table II. Every dimension is a named variable in one text file to which all parts link. No sketch dimension is typed as a number. Quantities that are not dimensions, such as the mainspring torque after 24 hours or the predicted balance amplitude, are variables of the same file, so they update with the geometry. This made it practical to revise the design each time a calculation required it.

III. GEAR TRAIN AND TOOTH FORM

A. Ratios

Each tooth of a 15-tooth escape wheel accounts for two beats, so the escape wheel must turn 16 times per minute. The second wheel must turn once per hour to carry a minute hand, which requires a ratio of exactly 60 between it and the fourth wheel. The chosen tooth counts (Table III) give

$$R = \frac{84}{12} \cdot \frac{60}{8} \cdot \frac{64}{8} \cdot \frac{96}{6} = 7 \times 7.5 \times 8 \times 16 = 6720, \quad (1)$$

and the rate closes exactly: $60 \times 16 \times 15 \times 2 = 28\,800$ beats per hour. The barrel turns once in 7 hours, so a 40-hour reserve needs 5.71 of the 8.75 turns the spring provides. The module falls from 0.140 to 0.065mm along the train because the torque falls at every stage, by a factor of 7 to 8 at the first three meshes and of 16 at the last.

TABLE III
MESHES OF THE GOING TRAIN

Mesh	Z/p	m (mm)	a (mm)	T (N mm)
Barrel / second	84/12	0.140	6.720	4.212
Second / third	60/8	0.120	4.080	0.602
Third / fourth	64/8	0.100	3.600	0.0802
Fourth / escape	96/6	0.065	3.315	0.0100

T : torque on the driving wheel at full wind, without losses.

B. NIHS 20-30 Profile

Watch pinions have 6 to 12 leaves and are driven by their wheels. With so few teeth an involute pinion is undercut, and the configuration lies outside the scope of the involute rating standard, ISO 6336 [3]. The teeth follow the Swiss standard for ogival gearing, NIHS 20-30 [4]. A wheel flank is a single circular arc of radius $2m$ from a pointed tip through the pitch point into a root fillet; a pinion flank is a radial line closed by a tip arc (Fig. 4).

The standard text is sold by the Swiss watch industry federation and was not available to the author. The eight profiles were generated with an open-source, third-party tool [5], and each was then rebuilt as a sketch whose only inputs are the standard's proportions, so that the teeth regenerate from the equations file.

For the wheel, with pitch, tip and root radii r_p , r_a , r_f and tooth thickness s^*m on the pitch circle, the flank is the arc of radius $R_o = 2m$ through the pitch point P and the tip T , and the root fillet centred on the gap centreline at height y_c follows from tangency:

$$y_c = \frac{|C|^2 - (R_o - r_f)^2}{2(C_y + R_o - r_f)}, \quad (2)$$

where C is the centre of the flank arc and C_y its component along the gap centreline. For a pinion of n leaves, the flank lies on a radial line at an angle $\alpha = p_l/n$ from the leaf centreline, where p_l is the flank factor of the standard, the tip arc of radius R_t is tangent to it at a distance

$$t = r_a \cos \alpha - \sqrt{2r_a R_t \sin \alpha - r_a^2 \sin^2 \alpha} \quad (3)$$

from the centre, and the root fillet has radius $\rho = y_c - r_f$ with $y_c = r_f/[1 - \sin(\pi/n - \alpha)]$. Only the standard's proportions enter as dimensions; everything else follows from tangency and coincidence relations.

The reconstruction matches the generated flanks, tips and pinion roots to better than $0.001\mu\text{m}$, and the wheel roots to within $1.5\mu\text{m}$ in a region no mating tooth reaches. The agreement is with the generator's output; that the generator implements the standard faithfully is taken on trust.

IV. CONTACT AND TOLERANCE ANALYSIS

The same tool rolls each pair of profiles through one pitch with a friction coefficient of 0.20 and reports efficiency, torque ripple and Hertz pressure. Its Monte Carlo mode repeats this with random errors of $\pm 0.015\text{mm}$ in centre distance and



Fig. 2. The thirteen unique parts, rendered from the SolidWorks models (not to a common scale). Wheels and barrel are brass, pinions, arbors and escapement parts hardened steel, the pallet stones and roller jewel ruby, the balance copper-beryllium, the hairspring Nivarox and the mainspring Nivaflex 45/5.

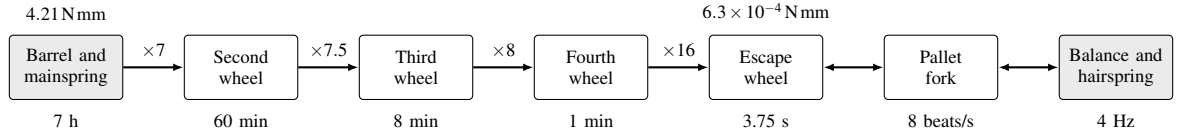


Fig. 3. Energy path of the mechanism. Speed multiplication of each mesh above the arrows, period of each arbor below, torques at full wind without losses. Between the escape wheel, the pallet fork and the balance the interaction is intermittent and acts in both directions.

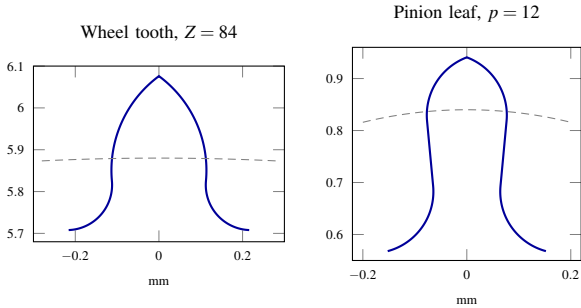


Fig. 4. NIHS 20-30 profiles of the first mesh ($m = 0.140\text{mm}$), plotted from the generated arc data. Dashed: pitch circle.

TABLE IV
CONTACT ANALYSIS, 1000 SAMPLES PER MESH AT FULL-WIND TORQUE.
THE BARREL MESH IS AT ITS ORIGINAL TOOTH WIDTH OF 0.25 mm.

Mesh	Efficiency	Torque ripple (%)		Hertz (MPa)
	mean (%)	mean	worst	
84/12	97.5	6.2	10.6	913
60/8	96.2	13.9	22.0	542
64/8	96.3	13.5	23.4	230
96/6	95.4	15.4	32.2	108

± 0.010 and $\pm 0.005\text{mm}$ in the wheel and pinion diameters. Fig. 5 shows the barrel mesh in the simulator, and Table IV gives the results, recomputed from the archived output.

Two results shaped the design. First, the barrel mesh failed. In line contact the maximum shear stress is $0.30 p_H$, so first yield occurs at $p_H = 1.67 R_{p0.2} \approx 670\text{MPa}$ for hard brass [6], and the simulated 913 MPa on teeth 0.25 mm wide exceeded it.

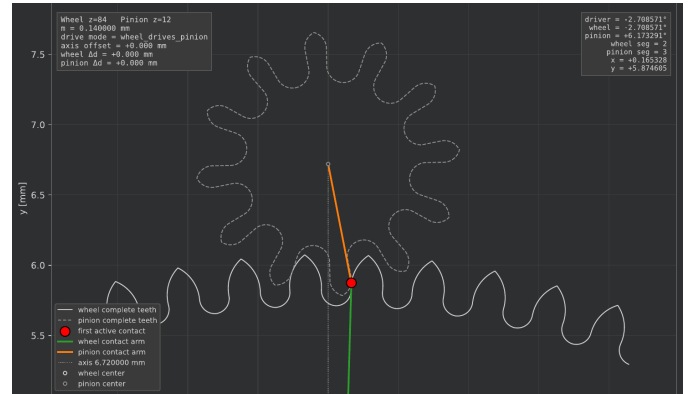


Fig. 5. The barrel mesh at the first position of active contact, as output by the contact simulator [5]: barrel teeth (solid), second pinion (dashed), contact point (red) and line of centres (dotted).

Since $p_H \propto b^{-1/2}$, the teeth were widened to 0.50 mm, giving 646 MPa. This value is a scaling of the simulated result; the analysis was not repeated at the new width.

Second, the centre distance dominates the running quality. Its correlation with torque ripple is 0.87 to 0.92 in every mesh (Fig. 6), against about 0.3 or less for the diameters. No sample collided or lost drive, so the tolerance is set by running quality: $\pm 0.010\text{mm}$ for the first three meshes and $\pm 0.005\text{mm}$ for the last, where it lowers the worst ripple from 32 % to 25 %. The mean mesh efficiencies multiply to 0.862.

V. SPRINGS AND OSCILLATOR

A. Mainspring

The spring is a strip of Nivaflex 45/5 ($E = 220\text{GPa}$ [7]), $0.095 \times 1.40\text{mm}$ in section. Filling half of the annulus between

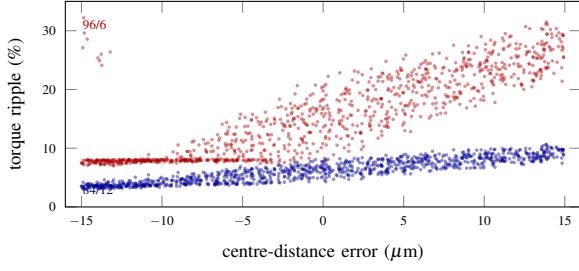


Fig. 6. Torque ripple against centre-distance error for the first and last meshes, 1000 Monte Carlo samples each.

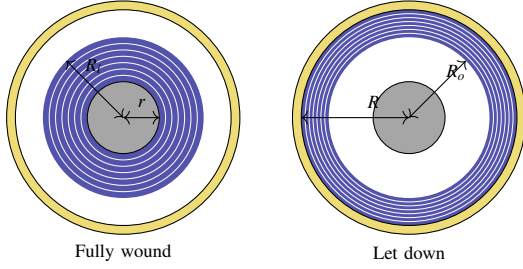


Fig. 7. Limiting states of the mainspring in the barrel, to scale ($r = 1.761$, $R = 5.282$, $R_i = R_o = 3.937$ mm). The difference between the coil counts, $22.9 - 14.2$, is the 8.75 turns of the barrel.

the arbor ($r = 1.761$ mm) and the drum wall ($R = 5.282$ mm) maximises the number of turns and fixes its length, $L = \pi(R^2 - r^2)/2t = 410$ mm, for 8.75 turns. Fig. 7 shows the two limiting states. Treated as a beam in pure bending, its stiffness and its largest moment are

$$K = \frac{Ebt^3}{12L} = 0.0537 \text{ Nmm/rad}, \quad (4)$$

$$M_{\max} = \frac{\sigma_d bt^2}{6} = 4.21 \text{ Nmm}, \quad (5)$$

with a design stress $\sigma_d = 2000$ MPa, and the torque after n turns is $M(n) = M_{\max} - 2\pi Kn$.

The first version of the model set $M_{\max} = 2\pi Kn$, which assumes a spring that is relaxed when let down. A real mainspring is preformed, so its full-wind torque is limited by stress and not by the number of turns. Correcting this, and checking the coils wound on the arbor, which would reach 5935MPa if the strip were straight when free, gave a required preset radius of at most 2.66mm for the inner coils.

B. Hairspring and Balance

The balance inertia from the CAD model is 3.59 mgcm^2 , carried in the equations as 3.60, and the rate requires 4Hz, so the hairspring stiffness is $k = 4\pi^2 If^2 = 2.274 \times 10^{-4} \text{ Nmm/rad}$. A Nivarox strip of 0.025×0.10 mm then needs a length $L = Ebt^3/12k = 108.8$ mm, or 11.35 turns, and is stressed to 126MPa at 330° . Because the rate varies as $t^{3/2}$, an error of 1% in thickness, a quarter of a micrometre, changes it by 1.5%, which is over 21 minutes per day. Each spring is therefore matched to its balance by adjusting its length, and

TABLE V
SENSITIVITY OF THE PREDICTED AMPLITUDE AND OF THE RUNNING TIME ABOVE 150° TO THE ASSUMED VALUES

Case	0 h	24 h	40 h	Hours
Nominal	252°	214°	185°	56.4
$Q = 200$	225°	192°	166°	48.6
$\eta_e = 0.25$	230°	196°	169°	50.2
Torque -5%	245°	207°	177°	52.0
All pessimistic	198°	169°	146°	37.3
All optimistic	318°	271°	234°	61.3

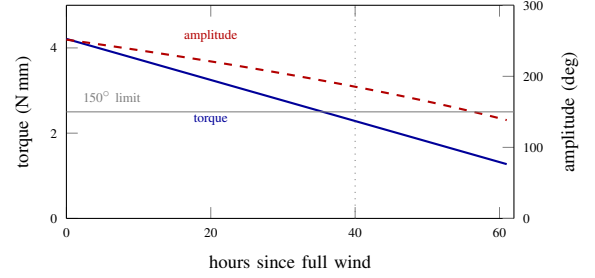


Fig. 8. Predicted barrel torque and balance amplitude over the run.

the value computed here is the nominal length from which that adjustment starts.

C. Amplitude

An oscillator of quality factor Q loses $2\pi/Q$ of its energy per period. In steady state the escapement supplies this from the work of the escape wheel over one tooth, $\phi = 2\pi/15$, with efficiency η_e :

$$\theta = \sqrt{\frac{Q\eta_e T_E \phi}{\pi k}}, \quad T_E = \frac{M\eta_t}{6720}. \quad (6)$$

With $Q = 250$, $\eta_e = 0.30$ [8] and $\eta_t = 0.70$, Fig. 8 shows the result. The first design, with a smaller barrel and the incorrect torque model, gave 196° at full wind and fell below the 150° running limit after 19 hours. The redesigned barrel (module $0.120 \rightarrow 0.140$ mm, spring height $1.00 \rightarrow 1.40$ mm) gives 252° , 214° and 185° at 0, 24 and 40 hours, and stays above 150° for 56 hours. The requirement is still met with $Q = 200$, the value quoted for an ordinary watch [9]. Table V gives the sensitivity to each assumption. Any single assumption at its unfavourable value leaves the 40-hour requirement met; only all three together fall just short. At the other extreme the full-wind amplitude reaches 318° , just above the limit of about 315° beyond which the roller jewel strikes the outside of the fork; the remedy would be a lower design stress in the mainspring.

VI. ESCAPEMENT

The escapement follows the classical construction for a 15-tooth wheel (Fig. 9). The stones span 60° , and the pallet axis lies where the tangents at the two locking points meet, at $R_e/\cos 30^\circ = 2.540$ mm from the wheel axis. The lever turns through 10.5° : 1.5° of lock, 8.5° of impulse shared between

TABLE VI
TOOTH AND LEAF BENDING ON THE NIHS PROFILES AT FULL-WIND
TORQUE, WITH SAFETY FACTORS IN FATIGUE AT THE TEN-YEAR CYCLE
COUNT

Mesh	F_t (N)	Wheel tooth		Pinion leaf	
		σ (MPa)	SF	σ (MPa)	SF
84/12	0.716	46	4.6	86	8.5
60/8	0.167	35	4.9	84	7.6
64/8	0.0251	6.3	22	15	36
96/6	0.0032	1.5	76	4.5	101

TABLE VII
PIVOT BENDING UNDER RUNNING LOAD AND UNDER A 5000g SHOCK
(HARDENED STEEL, $R_{p0.2} = 1600$ MPa, ENDURANCE 800MPa)

Arbor	d (mm)	Running		Shock	
		σ (MPa)	SF	σ (MPa)	SF
Second	0.12	198	4.0	303	5.3
Third	0.10	85	9.4	367	4.4
Fourth	0.08	33	24	524	3.1
Escape	0.07	4	> 100	383	4.2
Pallet	0.07	0		140	11.4
Balance	0.08	0		829	1.93

tooth and stone, and 0.5° of run to the banking. The lock is an overlap of only 0.033mm between tooth and stone, the most delicate dimension of the design. A draw of 12° exceeds the friction angle of 8.5° , so the lever is held against its banking pin, and a drop of 1.7° gives clearance at the cost of 14% of the wheel's rotation. The roller jewel radius follows from the lift angle of 52° by the sine rule, $r_j = 0.471$ mm.

VII. STRENGTH VERIFICATION

Each load-carrying element was checked under two load cases: the full-wind torque without losses, and a shock of 5000g. The shock level follows from a fall of 1m, which gives an impact velocity $v = \sqrt{2gh} = 4.43$ m/s, the velocity used in the ISO 1413 test [10]; brought to rest in 0.2mm, the deceleration is $v^2/2d = 5000$ g. It is an assumption about the stopping distance, at the upper end of the plausible range, and all shock stresses scale with it. The load is applied statically; for the balance on its arms, whose natural period of 0.31ms is comparable with the duration of the pulse, the dynamic response may be up to about 1.7 times higher.

Tooth bending cannot use tabulated form factors, since none exist for ogival teeth, so the section of highest stress was found by evaluating the cantilever stress $6F_t(r_L - r)/bs(r)^2$ along the real profile.

Table VI gives the tooth results. Pivots were treated as cantilevers, $\sigma = 32Fa/\pi d^3$; arbors by the Goodman criterion; wheel hubs by Lamé's equations for a press fit; the barrel lid and base as annular plates under the weight of the spring; and the locking impact by an energy balance on the bending tooth. Table VII gives the pivot results; the balance, the heaviest part relative to its pivots, is the critical one.

The plate check is typical of the method. The spring weighs 464mg, so at 5000g it loads the lid with 22.8N. Solving the axisymmetric plate equation $\nabla^4 w = q/D$ for an annulus with a free inner edge gives 1745MPa in a 0.10mm lid resting on a step, 436MPa at 0.20mm, and 158MPa at 0.20mm with the edge clamped. Thickness alone was therefore not enough, and the lid had to be both thickened and clamped by a groove. The press-fit check is similar in kind: the interference must exceed $1.32\mu\text{m}$ for the second wheel not to slip and stay below $1.68\mu\text{m}$ for its hub not to yield, a window no machining process can hold.

Table VIII summarises the outcome.

Two elements do not meet their criterion and are reported as such; a third, the mainspring, is deliberately designed outside the general criterion, as all mainsprings are, because any unused strength is unused energy. The barrel teeth have a margin of 1.04 against a required 1.2, and a width of 0.70mm is recorded for the next revision. The balance pivots need the shock-absorbing jewel settings that every modern watch has and that this model omits. Tooth bending is nowhere critical: the lowest fatigue factor is 4.6, and from the third mesh onwards the stresses are below 15MPa. Beyond the barrel, watch teeth are sized by what can be cut and not by what they carry.

Where stress does not govern, life is limited by wear, which was bracketed with Archard's law [11] for wear coefficients between 10^{-9} and 10^{-7} . The pivots lose less than $0.3\mu\text{m}$ in ten years even with degraded lubrication. The impulse faces of the escape teeth could lose $11\mu\text{m}$, a third of the lock, so the escapement sets the service interval, at five to seven years and not ten.

VIII. DESIGN REVISIONS

The first complete model was geometrically consistent and would not have worked. Table IX lists what the analysis found, in the order of discovery. Seven of the nine defects were corrected in the model; the wheel joints and the balance pivots are carried forward as requirements for manufacture. None of these defects was visible in the geometry; each appeared only when a number was calculated and compared with a limit.

IX. MOTION SIMULATION

A. Model and Settings

The mechanism was run in SolidWorks Motion, a rigid multibody solver with contact, as three studies: a kinematic check of the train, the escapement alone, and the complete train driven by a torque on the barrel. The hairspring is a torsional spring with a damper representing the losses of the balance; the gear meshes are kinematic constraints, with their losses applied to the drive torque. Contact is modelled only where it governs the behaviour: teeth on stones, fork on roller jewel, and lever on banking pins.

The simulator imposed one practical limit. It becomes unstable on the repeated impacts of an escapement, and no run could be carried much beyond a second of simulated time. The studies of the escapement and of the train are therefore

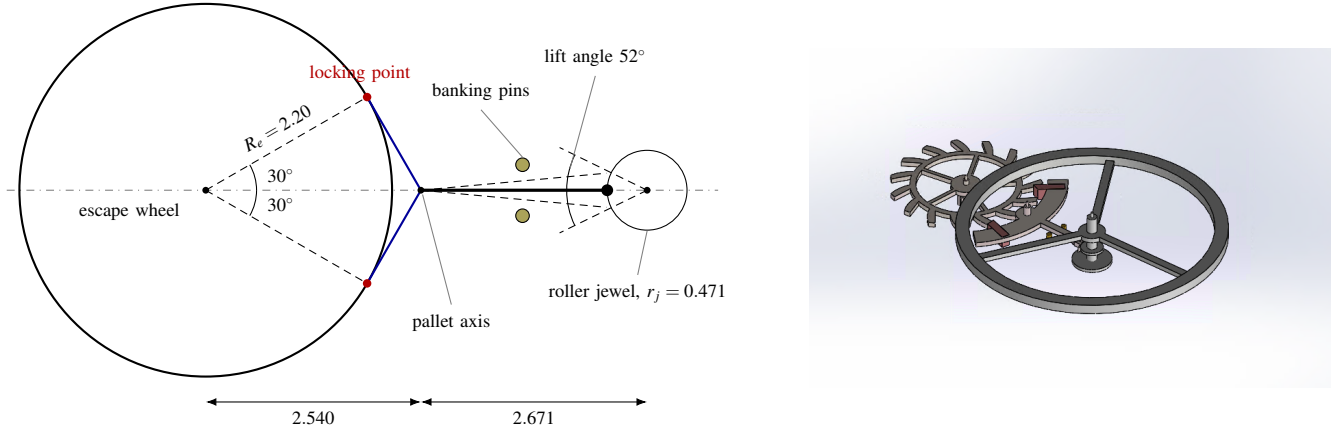


Fig. 9. Left: layout of the escapement to scale (mm), with the lever in its mid position and against each banking pin. Right: the escapement and balance as modelled.

TABLE VIII
GOVERNING CHECK, MARGIN AND CRITERION FOR EACH ELEMENT

Element	Governing check	Margin	Required	Outcome
Barrel teeth	contact pressure, full wind	1.04	1.2	not met at 0.50mm width; 0.67mm required
Other teeth and leaves	bending fatigue	≥ 4.6	1.5	met; sized by manufacture, not by load
Train, escape, pallet pivots	bending under shock	≥ 3.1	2.0	met
Balance pivots	bending under shock	1.93	2.0	not met; shock-absorbing settings required
Arbors	bending and torsion, fatigue	≥ 11	1.5	met
Wheel-to-arbor joints	press-fit interference window	$0.36 \mu\text{m}$		not manufacturable; wheels to be riveted
Spokes and balance arms	bending under shock	≥ 1.51	1.5	met
Barrel lid, base	plate bending under shock	2.53, 2.05	1.5	met after thickening, with the lid clamped
Mainspring	bending at full wind; eye	1.24; 1.10	by design	below the general criterion as intended; eye to be verified by finite elements
Hairspring	fatigue at the pinned ends	≥ 1.8	1.5	met, on an estimated endurance strength
Escape teeth	locking impact	1.6 to 2.2	1.2	met for a corner radius of 0.005 to 0.010mm

TABLE IX
DEFECTS FOUND BY ANALYSIS AND THE CHANGES THAT RESOLVED THEM

Subject	Finding	Change
Power source	Predicted amplitude 196° at full wind, below 150° after about 19 hours instead of 40	Barrel module $0.120 \rightarrow 0.140\text{mm}$, spring height $1.00 \rightarrow 1.40\text{mm}$; 185° at 40 hours
Mainspring model	Torque law implied zero torque when let down; coils on the arbor unchecked (5935MPa if unset)	Stress-capped torque model; preset radius $\leq 2.66\text{mm}$ for the inner coils
Time display	A second wheel of 54 teeth turned once in 54 minutes, so no arbor could carry a minute hand	$Z_2 = 54 \rightarrow 60$
Tooth form	One estimated set of proportions was shared by all meshes, with dedendums deeper than the standard	One NIHS 20-30 set per mesh, taken from the generated profiles
Barrel teeth	Contact pressure 913MPa on 0.25mm wide brass teeth, above first yield at 670MPa	Width $0.25 \rightarrow 0.50\text{mm}$ (646MPa); 0.70mm to meet the full criterion
Second pivot	The stronger spring raised the running load on the second arbor by 59%	Pivot diameter $0.10 \rightarrow 0.12\text{mm}$
Barrel lid	A 0.10mm lid resting on a step reaches 1745MPa under shock, a safety factor of 0.23	Lid 0.20mm, snapped into a groove (2.53); base $0.15 \rightarrow 0.18\text{mm}$
Wheel joints	A press fit of the second wheel needs an interference between 1.32 and $1.68 \mu\text{m}$	Wheels riveted on a pinion shoulder
Balance pivots	At 5000g the pivots reach 829MPa, a safety factor of 1.93 with rigid bearings	Not resolved in the model; shock-absorbing jewel settings are required

short, about eleven beats, which is enough to show that the mechanism runs correctly but not to reach a settled amplitude.

The default settings failed, and each replacement was de-

rived from the physics:

- *Integrator*: The default assumes a slowly varying step. At each impact the step must fall from $10 \mu\text{s}$ to below $1 \mu\text{s}$,

TABLE X
PRINCIPAL SOLVER AND CONTACT SETTINGS

Setting	Value
Integrator	WSTIFF, 50 iterations
Step: maximum, initial, minimum	10, 1, 0.1 μ s
Accuracy	10^{-4} , precise contact
Contact stiffness, exponent	10^4 N/mm, 1.5
Contact damping, penetration	10^{-3} Ns/mm, 0.5 μ m
Friction (specified)	0.12, 0.15, 0.20, kinetic only
Hairspring	3.97×10^{-6} Nmm/deg with damper
Drive	2.274 Nmm on the barrel

and only the WSTIFF integrator, which recomputes its coefficients when the step changes, ran through.

- *Step size.* A landing tooth moves at about 70mm/s. A maximum step of 10 μ s limits its travel between evaluations to 0.7 μ m, comparable to the contact penetration.
- *Contact.* Library values intended for machine-sized parts allow a penetration of 0.1mm, three times the lock. A stiffness of 10^4 N/mm with exponent 1.5 gives a penetration of about 0.3 μ m, and the damping was set to about half the critical value for a tooth.

Table X lists the settings. The simulation was brought up in stages, each with a known expected result, so that a failure could be attributed to one cause: the spring alone, which must give a period of 0.25 s; then the fork and banking pins, with the jewel reaching the fork 0.055 s after release; then the torque and the tooth contact, where the lever must stay on its pin under the draw; and finally the full run.

B. Results

In the kinematic study, with the fourth wheel driven at 1 rev/min for 11 s, the fourth, third and second wheels and the barrel turned at -6.000 , $+0.750$, -0.100 and $+0.01429^\circ$ per second, which are the values required by the tooth counts to the precision of the output, with the directions alternating along the train. Fig. 10 shows the complete train over 1.4 s, eleven beats, with the balance released 140° from rest; a recording is available [12]. Table XI compares measurement with design.

The period and the ratios confirm the oscillator and the train. Three further results follow from the data.

1) *Rate:* The period is short of the design value by 0.095 %, a gain of about 82 seconds per day, and most of this lies in the oscillator and not in the escapement: in the free arcs the balance runs at 4.004 Hz, because the inertia of the CAD balance, 3.59 mg cm², is slightly below the 3.60 for which the hairspring was sized. A real watch removes such an error by regulation.

2) *Lock and Drop:* The escape wheel advances in alternate steps of 12.29° and 11.71° , which shows that the two stones do not lock exactly half a pitch apart. The recoil of the wheel at unlocking tells the same story from the locking side. Its mean, 0.19° , agrees with the 0.18° expected from the lock

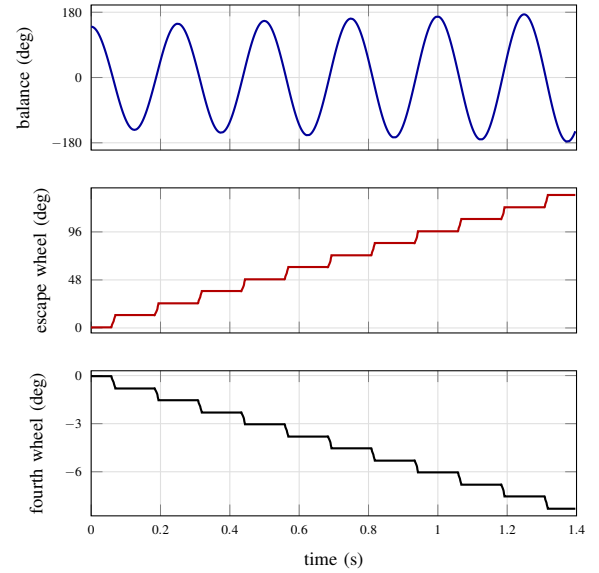


Fig. 10. Measured response of the complete train driven from the barrel: balance angle about rest, escape wheel and fourth wheel (seconds hand).

TABLE XI
COMPLETE TRAIN: MEASURED AGAINST DESIGN

Quantity	Design	Measured
Balance period	0.2500 s	0.2498 s
Natural frequency, free arcs	4.000 Hz	4.004 Hz
Wheel while locked	stationary	stationary to 10^{-5} deg
Recoil at unlocking	0.18°	0.27° and 0.12° alternately
Beat rate	8 per s	8.01 per s
Ratios to escape wheel	16, 128, 960	exact to 10^{-7}
Fourth wheel	6° per s	6.009° per s
Escape wheel step	12°	12.29° and 11.71° alternately
Balance amplitude	221° settled	146° rising to 175° ; not settled

depth and the draw angle, but the two values imply a lock of about 2.2° on one stone and 0.9° on the other against a design value of 1.5° . The positions of the stones or of the banking pins need correction.

3) *Energy Delivered:* The energy delivered by the escapement can be measured. Between impulses the balance is free, and its energy $\frac{1}{2}I\dot{\theta}^2 + \frac{1}{2}k\theta^2$ changes only by damping; the step in this energy from one free arc to the next is the energy delivered by one impulse. Against the work of the escape wheel in one beat,

$$W_{\text{beat}} = T_E \frac{\Phi}{2} = 7.09 \times 10^{-5} \text{ Nmm}, \quad (7)$$

the escapement delivers a fraction of 0.59 to 0.64, after allowing for the decay within each arc. That decay corresponds to $Q \approx 420$ and not to the specified 250, which indicates that the damping coefficient, a number of order 10^{-10} , was rounded when it was entered.

This run was made without contact friction, so the figure is the efficiency of the geometry with drop and impact losses

TABLE XII
BALANCE AMPLITUDE OVER SUCCESSIVE HALF SWINGS OF THE
COMPLETE-TRAIN RUN (DEGREES)

1	2	3	4	5	6	7	8	9	10
146.4	150.3	154.1	157.8	161.0	164.1	167.0	169.8	172.7	175.4

TABLE XIII
PRINCIPAL FUNCTIONAL TOLERANCES AND THEIR ORIGIN

Feature	Tolerance	Origin
Centre distance, meshes 1 to 3	± 0.010 mm	Monte Carlo study
Centre distance, mesh 4	± 0.005 mm	ripple 32 $\rightarrow$ 25%
Escapement centres	± 0.005 mm	changes the lock by $\pm 0.23^\circ$
Guard dart to roller	0.020 ± 0.010 mm	must stay positive
Clearance under balance rim	≥ 0.12 mm	rim deflection under shock
Locking corner radius	0.005 to 0.010 mm	impact pressure and lock depth

only, and it is consistent with the 0.30 assumed for a real escapement once friction is included. The amplitude was still rising by about 3° per beat when the run ended (Table XII), for the reason given above. The approach to a steady amplitude has a time constant of the order of $Q/\omega \approx 10$ s, so the settled value cannot be read from this run.

X. MANUFACTURABILITY

The thirteen modelled parts correspond to between 25 and 30 manufactured components, since each wheel is in reality a brass wheel riveted to a steel pinion, and the pallet fork and balance are assemblies. The analysis fixes several manufacturing decisions. The wheels must be riveted. The barrel lid must be clamped by its groove. The barrel brass must be in a hard temper, since $R_{p0.2} = 400$ MPa is consistent with hard-rolled strip (temper R490 of EN 1652, tabulated for the closely related CuZn39Pb3 [13]) but not with soft bar. The locking corner of the escape teeth must be 0.005 to 0.010 mm and polished. Each mesh needs its own cutter, because the NIHS proportions depend on the pinion leaf count. Both springs are bought to a functional specification, which this work supplies: a torque of 4.21 Nmm at full wind and at least 1.26 Nmm when let down for the mainspring, and a stiffness of 2.274×10^{-4} Nmm/rad for the hairspring. The escape wheel and lever, flat parts with features of 5 to 50 μ m, are natural candidates for LIGA or etched silicon. Table XIII lists the tolerances that follow from the analysis.

XI. LIMITATIONS AND FUTURE WORK

Nothing has been built or measured. The frame, the bearings and their clearances are not designed, so the tolerance values are requirements on parts that do not yet exist. The springs rest on beam theory; the mainspring torque curve and the isochronism of the hairspring need nonlinear finite element

analysis. The efficiencies and the quality factor in (6) are literature values, and the simulation ran without contact friction and for too short a time to give a settled amplitude. The NIHS standard text was not consulted, and the endurance strengths of the alloys at 10^8 to 10^9 cycles are estimates. No compliance with a standard is claimed: ISO 1413 is not met as modelled, and the chronometer and magnetic tests of ISO 3159 [14] and ISO 764 [15] apply only to a finished watch.

The next steps follow directly: widen the barrel teeth to 0.70 mm; correct the stone positions to equalise the drop; find the settled amplitude by releasing the balance from several angles in short runs with friction applied; analyse both springs by finite elements; and design the frame with jewelled bearings and shock settings.

XII. CONCLUSION

A mechanical watch mechanism of thirteen parts was designed from its requirements, with every dimension traceable to a formula. The tooth form reproduces the generated standard profiles on every working surface, and a tolerance study set the limits within which the gears must be made. Verification from first principles exposed nine defects, and without it the model would have assembled without error and run for less than a day. A contact simulation of the complete train confirmed the rate and the transmission ratios, and revealed an asymmetry in the escapement that the calculations had not predicted. The derivations, the worked calculations for every result quoted here and the full list of variables are in the companion report [2].

ACKNOWLEDGMENT

The tooth profiles and the contact study were produced with the Watchmaking Gear Simulator by hdrpano [5]. That software is third-party work and is gratefully acknowledged. The concept follows Ciechanowski's article [1], which was used by reading it; no model or code from it is reproduced here.

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